Exp 11:

Routing a Network layer

a)

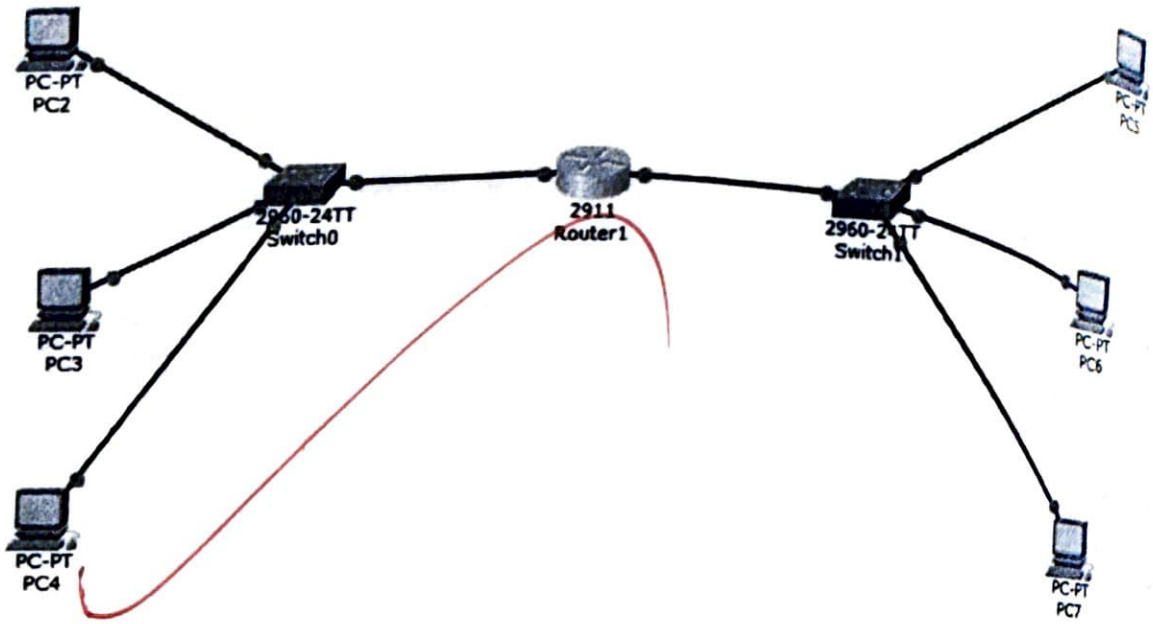
Aim: To simulate and configure static routing protocol using cisco packet Tracer and observe packet routing between different networks.

Procedure

a) Static Routing

1. Open Cisco Packet Tracer
2. Place three routers, two switches and four PCs
3. Connect the devices using appropriate ethernet cables
4. Configure the IP addresses of the router interface and PCs
5. Enable the router interfaces using the no shutdown command
6. Configure static routes on the routers using the ip route command
7. Verify the routing table using show ip route static
8. Test connectivity between PCs using the ping command
9. Use
   Tracert <destination-IP>

Result: Thus, Static routing was successfully simulated and packet Tracer and packet transmission Configured backup routes was verified


PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
2911
Router1
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6
PC-PT
PC7

9 exp 11.b Simulate RIP using Cisco Packet Tracer

Aim:

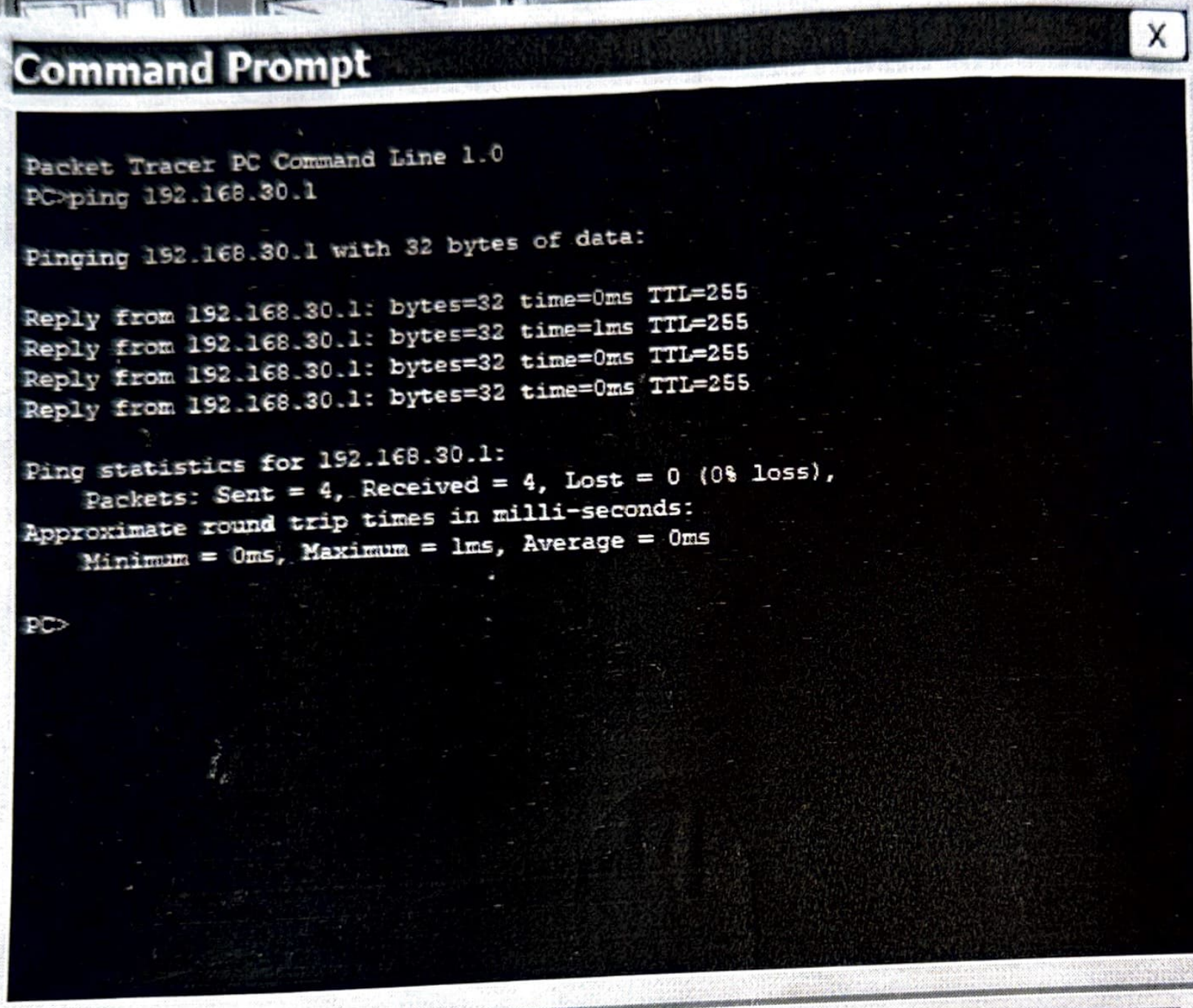
To configure and simulate the routing information protocol (RIP) between routers using Cisco packet tracer and verify network connectivity

Steps:

1) Open Cisco packet tracer
2) place 2 or 3 routers, switches and PCs
3) Connect the devices using Ethernet cable
4) Assign IP address to PCs and router interfaces
5) Enable all router interface using no shutdown
6) Configure the RIP on the routers
7) Configure the appropriate network address on each routers according to the topology
8) Check the routing table using: show ip route
9) Use the ping command to test communication
9) Use the ping command to test communication between PCs in different networks.
10) Use simulation mode to observe how RIP learn and select routers

Result:

RIP was successfully configured between the routers using Cisco packet tracer. The routers learned the required network routes dynamically and communication between different network was successfully verified


Command Prompt
X
Packet Tracer PC Command Line 1.0
PC>ping 192.168.30.1
Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Ping statistics for 192.168.30.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>

PC-PT
PC1